



Brief.

"You should enjoy the little detours to the fullest. Because that's where you'll find the things more important than what you want."





My philosophy has always been that "you should enjoy the little detours... because that's where you'll find the things more important than what you want." So, instead of a rigid map, I started taking detours. One of those detours led me to code in my second year, and it changed everything.

STORY TIME

They tell you to have a five-year plan. I just wanted to make it through the first week. Dropping into SRMIST from a small town felt less like a plan and more like a hard landing on a new planet.

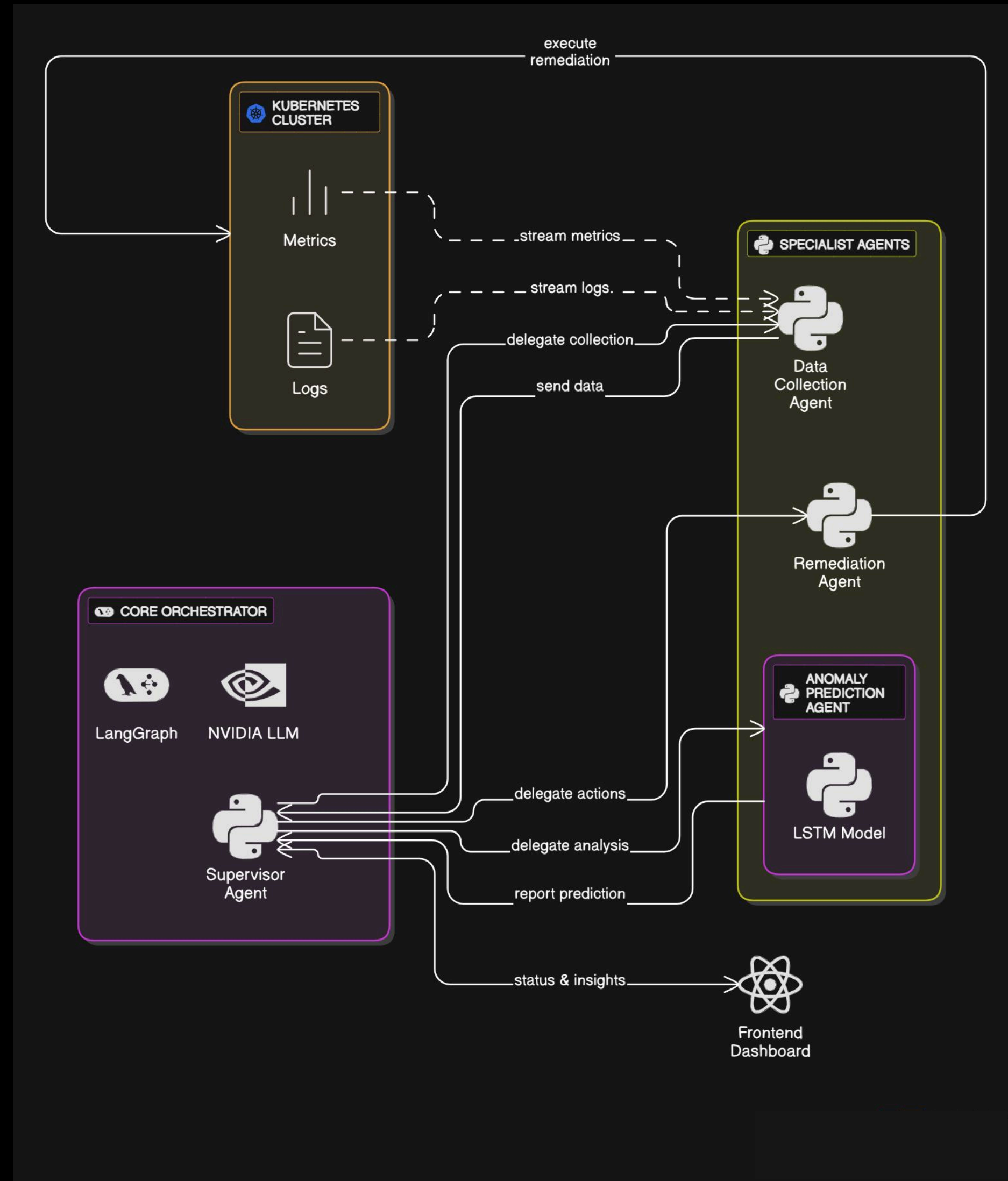
My strategy was simple: "Whatever happens, happens". But that isn't a passive statement; it's a challenge. It's the freedom to chase any signal, build any system, and dive into any problem without a map, just to see if you can. It's a strategy that led me to places I never planned: the final round of national coding contests, the helm of a startup, the heart of product design. This portfolio isn't a map of where I was going. It's the map I drew after the territory had been explored.

PRESENT

That single detour turned feeling inadequate into an insatiable curiosity. It pushed me to see challenges not as roadblocks, but as opportunities to create.

- This curiosity led me to design, shaping products for real users and learning to see through their eyes¹.
- It pushed me to build systems that work under pressure, competing against teams with years more experience and somehow holding my own.
- It gave me the business sense to see market gaps, leading me to found my own ventures.

I've learned that my strength isn't in specializing in one thing, but in understanding how all the pieces connect. Today, I'm not just a developer, a designer, or an entrepreneur. I am a multifaceted creator, driven to build products, systems, and a legacy that lasts.



Project Description:

This system was designed for the Guidewire DEVTrails national competition, with the goal of creating an autonomous, enterprise-grade platform for Kubernetes monitoring and remediation. As the lead architect, my primary responsibility was to design and implement the hierarchical, multi-agent system shown here. My core contributions were focused on building the system's central intelligence and workflow.

- The Supervisor Agent: I developed the core orchestrator using LangGraph to manage the state and workflow between all specialist agents. It leverages an NVIDIA LLM for complex, context-aware decision-making and dynamic task delegation.
- The Specialist Agents: I designed the logic for the three distinct worker agents:
 - a. A Data Collection Agent for gathering real-time metrics and logs.
 - b. An Anomaly Prediction Agent that uses a powerful LSTM model to achieve 94% detection accuracy.
 - c. A Remediation Agent that executes targeted actions—like pod restarts or resource scaling—directly on the cluster.

The Workflow: I orchestrated the entire flow visible in the diagram. The Supervisor delegates data collection, receives the data, passes it to the prediction agent, receives the anomaly report, and then delegates the appropriate remediation action. This closed-loop, autonomous design is what made our solution so resilient and effective.



K8s Anomaly Detection & Remediation UI Brief:
This dashboard visually simulates the monitoring and remediation processes of Kubernetes clusters, displaying real-time metrics, anomaly detection results, and remediation actions in an intuitive format.

Important Note: At this stage, the UI is purely simulated and not connected to any live Kubernetes cluster. It serves as a proof of concept to demonstrate how the system’s outputs and workflows would be presented to users in a real-world scenario.

Project Type: Group Project for Guidewire DEVTrails 2025 National Competition

My Role: Team Lead | Architect | DevOps

Impact:

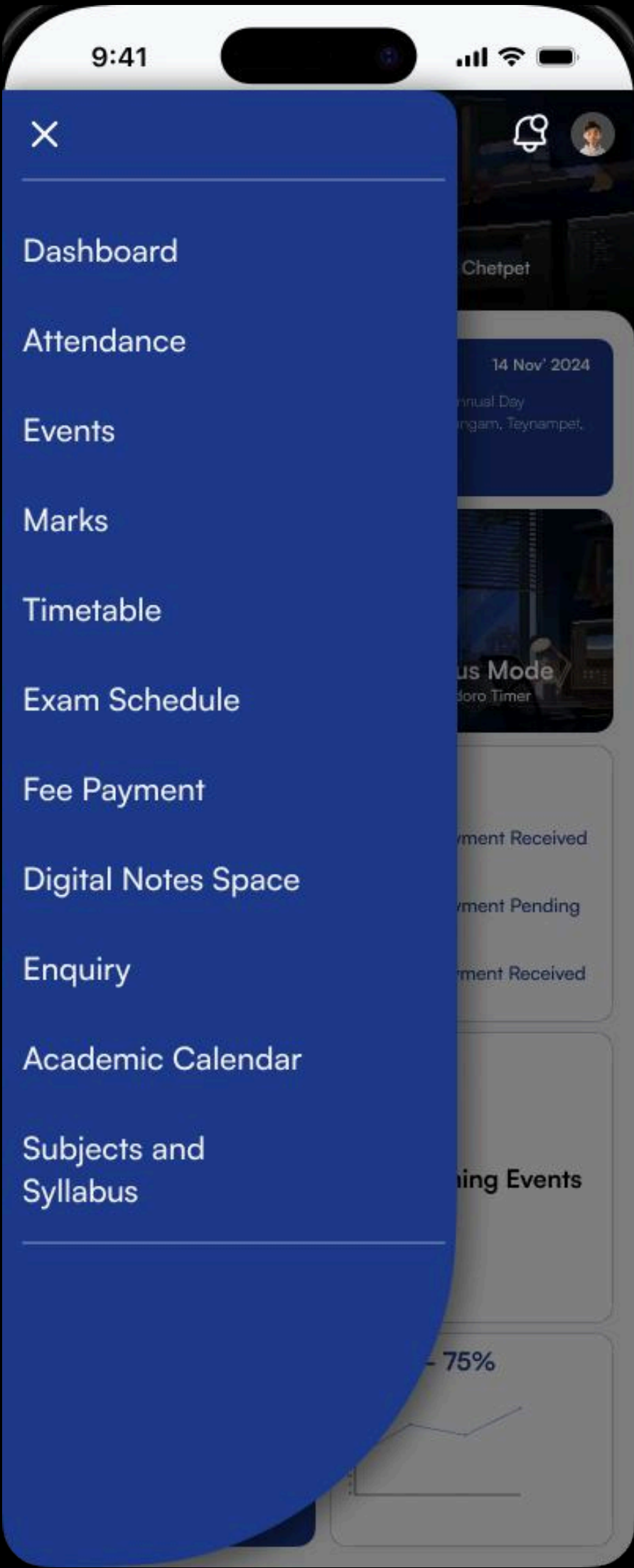
- Our system achieved a Top 5 Finalist position out of 739 competing teams, validating the technical excellence and innovation of our multi-agent approach.
- This result earned us a main-stage presentation slot at the Guidewire DEVSummit 2025, where we pitched our solution live to the Guidewire CTO, the PwC CEO, and an audience of 1,500 tech leaders at the Sheraton Grand, Bengaluru.

What I Learned

This project was a masterclass in two areas. Technically, I learned to architect truly autonomous systems using cutting-edge tools like LangGraph. Professionally, pitching our deeply technical solution to a C-suite audience taught me how to translate complex architecture into a compelling business case under extreme pressure.

Supporting Media Links: https://docs.google.com/document/d/19CnV3FAKfxVoINc69C8G4a-0levJ_8qQbBPYxDaz3rU/edit?usp=sharing





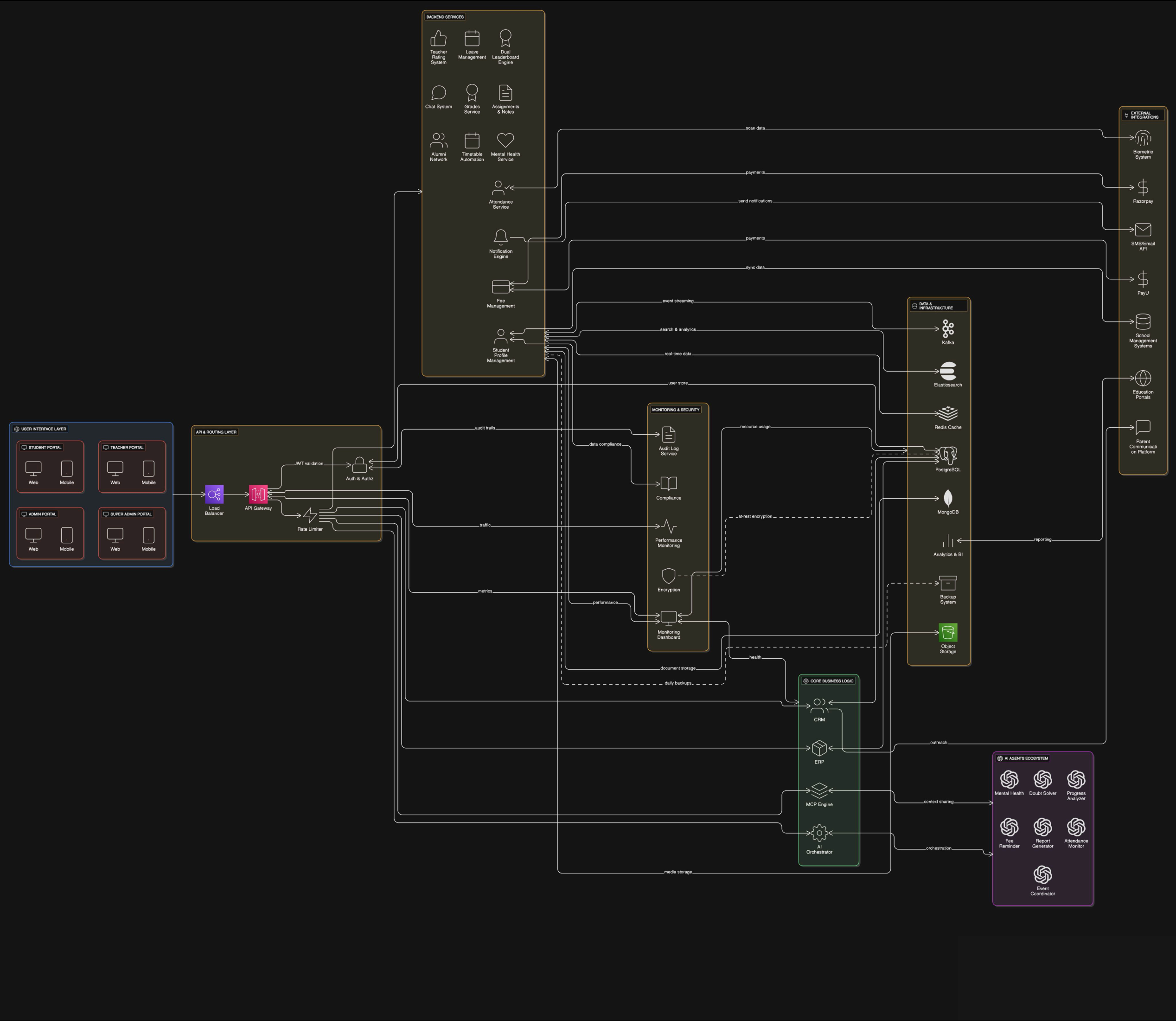
Project Description

This venture addresses a critical challenge in the Indian education market: the student isolation and intense pressure caused by overcrowded classrooms (often 50+ students). As Founder and Technical Lead, I am building a comprehensive SaaS platform to solve this, starting with a 10,000-student pilot at my own school.

The architecture is revolutionary, representing the first educational platform to unify CRM, ERP, Model Context Protocol (MCP), and a multi-agent AI system. This creates an intelligent, proactive ecosystem designed to support students, empower teachers, and streamline administration.

My Core Innovations:

- **Dual Leaderboard System:** To combat unhealthy competition, I designed a system with a traditional leaderboard for top performers and a unique Progress Leaderboard that ranks students by their own academic improvement, celebrating personal growth²³.
- **AI-Powered Mental Health Monitoring:** The system includes an agent that detects significant drops in a student's performance and discreetly alerts teachers, enabling early, compassionate intervention.
- **Context-Aware Intelligence:** Leveraging an MCP Engine, the platform understands the full context of a student's journey—academic history, learning patterns, and emotional state—allowing for truly personalized support.
- **Multi-Agent Automation:** I designed a hierarchical system of specialist AI agents (for doubt-solving, attendance, reporting) orchestrated by a central supervisor, moving beyond simple chatbots to true, autonomous institutional management.



Why This Architecture is Revolutionary for the EdTech Market

First-of-its-Kind "System of Systems" Integration

While competitors offer isolated solutions, my platform is the first to cohesively unify four distinct domains into a single, intelligent ecosystem:

- CRM (Customer Relationship Management): For student and parent engagement.
- ERP (Enterprise Resource Planning): For institutional administration and resource management.
- MCP (Model Context Protocol): As a central intelligence layer for real-time contextual awareness².
- Multi-Agent AI: For autonomous, proactive task execution¹.

This integration creates a system where every component enhances the others. For example, ERP data on student attendance can inform the CRM's communication with parents, all orchestrated with the contextual understanding provided by the MCP engine.

In conclusion, my architecture is revolutionary because it doesn't just digitize existing school processes. It reimagines the entire educational ecosystem by creating a proactive, intelligent, and empathetic platform.

Impact

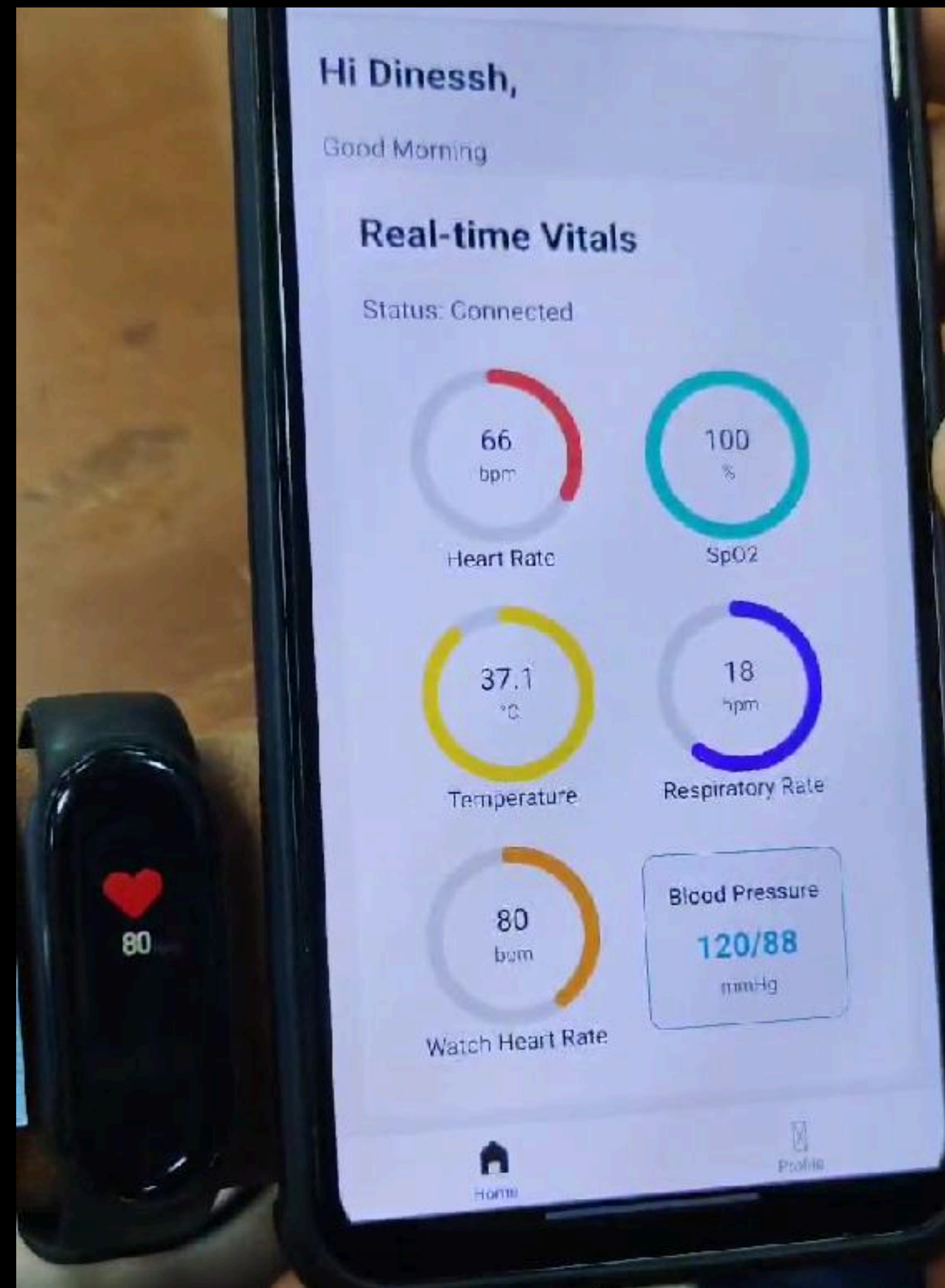
- **Market Disruption:** Targeting the underserved Indian EdTech market by offering a holistic management solution that prioritizes student wellbeing—a significant departure from competitors that focus only on content delivery or basic administration⁴⁵.
- **Scalable Vision:** Architected as a multi-tenant SaaS platform, Student Space is designed for rapid deployment across thousands of schools in the region, making advanced educational management accessible to all³.
- **Real-World Validation:** A functional live platform and a 10,000-student pilot at my own school demonstrate the project's viability and real-world application

My Role & Contribution

As the Founder & Technical Lead, I am responsible for the entire project lifecycle. I identified the market gap, architected the comprehensive technical solution from the ground up, and am leading the hands-on development and team coordination to bring this vision to life.

What I Learned

This entrepreneurial journey has been a masterclass in building a product from the ground up. I learned to translate a deep understanding of a real-world problem into a complex, cutting-edge technical architecture, moving beyond simple digitization to true intelligence augmentation. It taught me to balance ambitious features like AI-driven monitoring with human-centric values like student mental health, and to design systems that don't just solve problems, but transform the entire user experience.



Project Type: Group Project – CodeFest 2025 (24-Hour Inter-College Hackathon)

Project Brief: RPM – Remote Patient Monitoring

In a 24-hour hackathon, our team faced a project-killing problem: we had no hardware to monitor a patient's heart rate. With the clock ticking, I took a calculated risk. I disassembled my own Fitbit—with zero prior electronics experience—and successfully isolated its optical heart rate sensor.

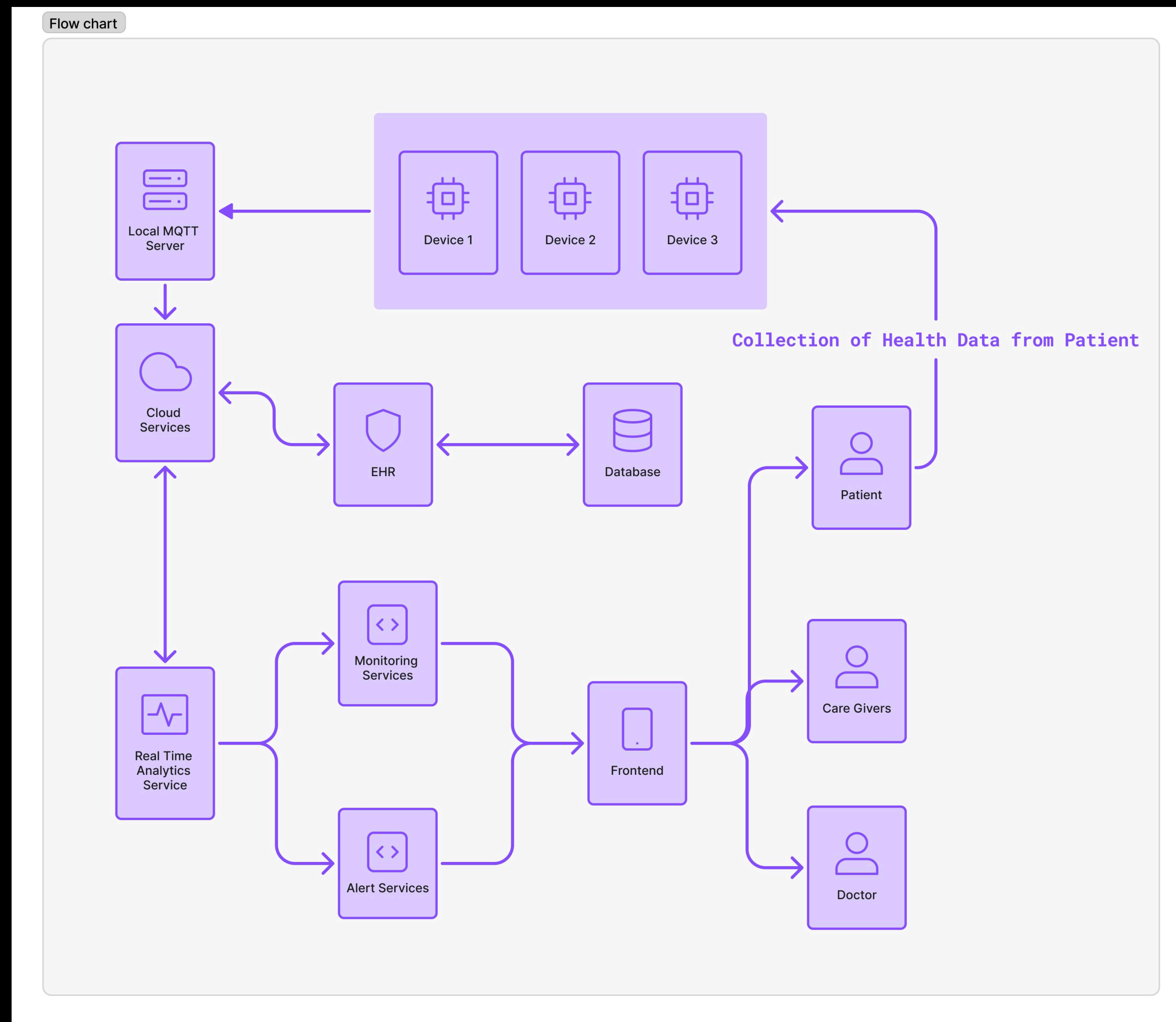
From that single component, our team built a complete, end-to-end monitoring platform. We secured 3rd Place in a highly competitive inter-college hackathon, demonstrating our ability to overcome critical obstacles with hands-on creativity and deliver a functional, award-winning product under extreme pressure

This combination of on-the-fly hardware hacking and robust full-stack development was instrumental to our 3rd Place finish.

This properly captures the technical audacity and skill of what you accomplished in 24 hours.

My Role: Full Stack JS Developer | Hardware Hacking

Supporting Media: <https://docs.google.com/document/d/1lQAZVH0kN5fIGK3E7pplyYdSh-WeL85acwSVk0QtTGw/edit?usp=sharing>



Key Architectural Strengths

This architecture is not just a simple data pipeline; its design has several inherent strengths:

- **Scalability:** By using cloud services and a microservices approach (separating Analytics, Alerting, etc.), the system can easily scale to handle thousands or even millions of patients without a decline in performance.
- **Real-Time Capability:** The use of MQTT and real-time analytics ensures that critical health information is processed and delivered with minimal latency, which is essential for emergency situations.
- **Modularity and Separation of Concerns:** Each component (MQTT, EHR, Analytics) has a distinct responsibility. This makes the system easier to develop, maintain, and upgrade. For example, a new type of analysis can be added to the analytics service without affecting the data collection process.
- **Security & Compliance:** The explicit inclusion of an EHR component signals a design that prioritizes data security and privacy, a non-negotiable requirement in the healthcare industry.
- **Reliability:** MQTT is designed for reliable message delivery, ensuring that data from patient devices is not lost, even with unreliable network conditions.

RPM System – How It Works: A Simple 3–Step Process

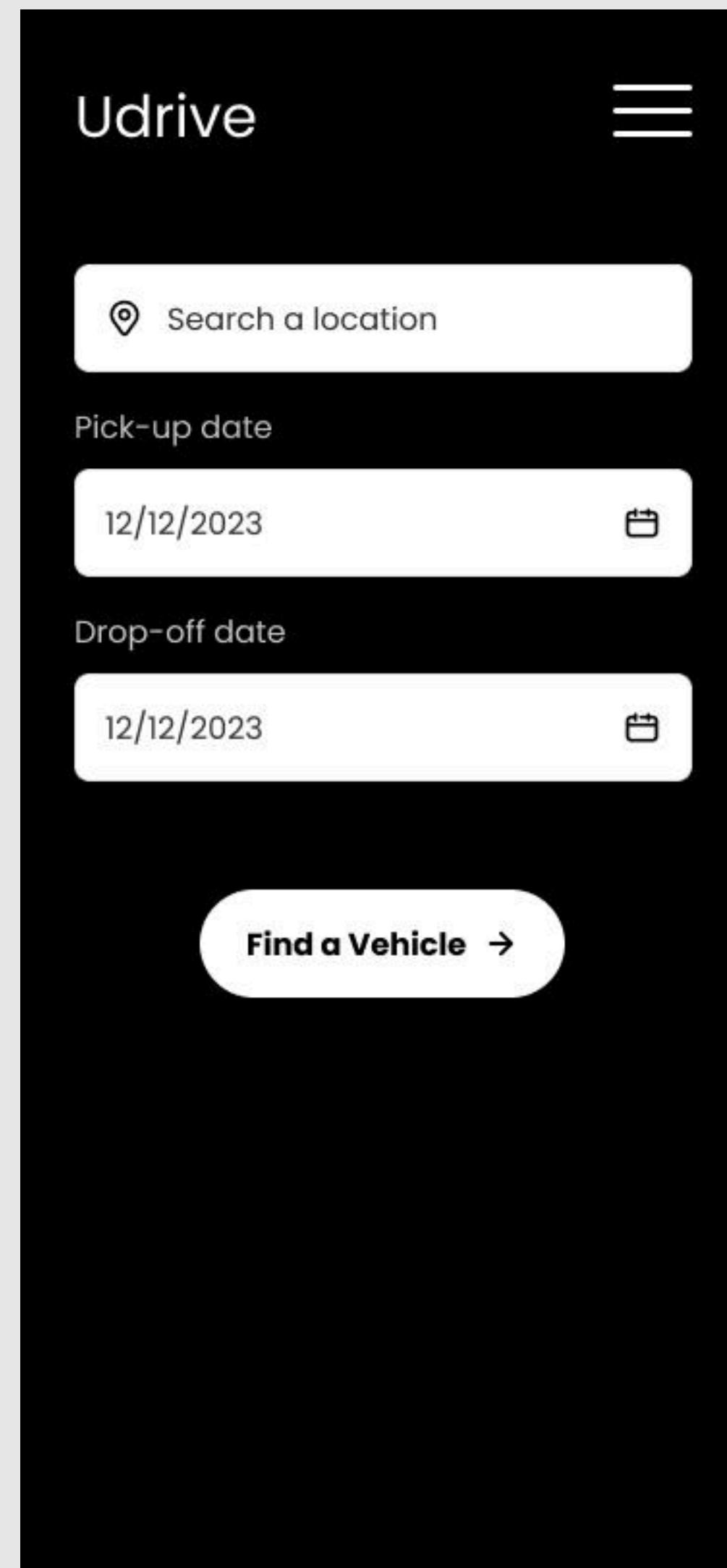
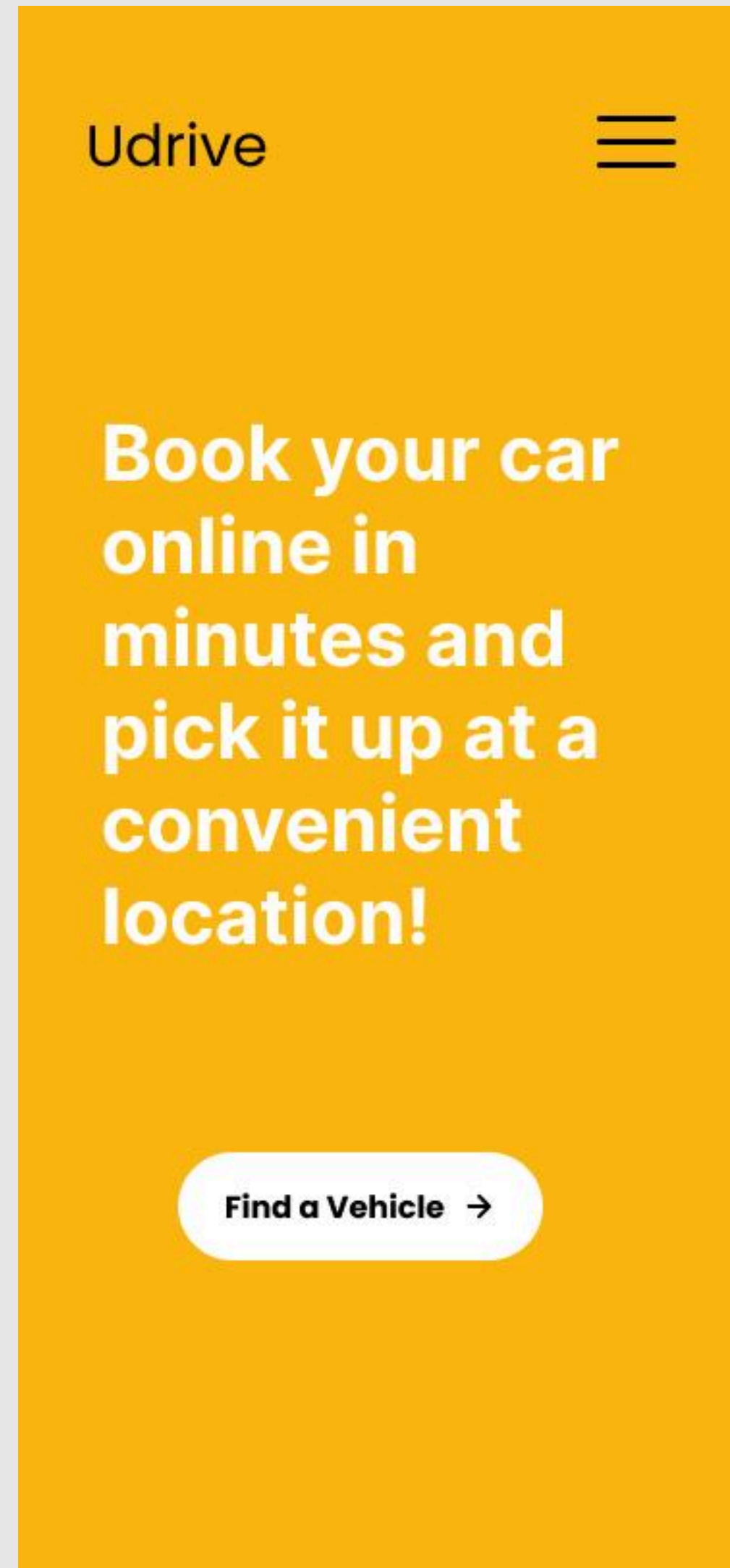
Our Remote Patient Monitoring (RPM) system operates on a continuous, real–time loop designed for immediate action.

1. **Data Collection:** A wearable sensor, like the heart rate sensor I extracted from a Fitbit, continuously captures a patient's vital signs (e.g., heart rate, O2 levels)4.
2. **Real–Time Transmission:** The sensor transmits this data instantly to a central cloud platform using efficient IoT protocols like MQTT. This ensures that even with low power or intermittent connectivity, the data is delivered reliably45.
3. **Analysis & Alerting:** The platform's backend immediately analyzes the data stream. If any vital sign crosses a predefined critical threshold, an automatic alert is pushed to a doctor's dashboard, enabling them to see the problem in real–time

Impact: Why Remote Patient Monitoring is Necessary?

Remote Patient Monitoring (RPM) is not a fancy tool; it's a critical shift in how modern healthcare is delivered, with a proven impact on both patient lives and the healthcare system. Its necessity is demonstrated in three key areas:

- **It Transforms Healthcare from Reactive to Proactive:** RPM allows doctors to identify a patient's deteriorating condition before it becomes a crisis. By continuously monitoring vital signs, providers can prevent emergencies instead of just reacting to them, which is especially critical for managing chronic diseases like heart failure and COPD.
- **It Reduces the Strain on the Healthcare System:** RPM is consistently shown to lower healthcare costs and alleviate the burden on overworked clinical staff. Studies prove it leads to fewer emergency room visits, prevents costly hospital readmissions, and shortens hospital stay
- **It Provides a Clearer, More Accurate Health Picture:** A brief office visit provides only a snapshot of a patient's health. RPM provides the full movie. By giving doctors access to continuous, real–world data, it leads to more accurate diagnoses, better–informed treatment plans, and greater patient engagement in their own care



Project4: Udrive – Car Rental Platform

Project Type: Professional Experience (First Internship)

My Role: Product Designer & Mobile Developer

Project Brief:

Udrive is a cross-platform car rental application designed to provide a seamless and intuitive booking experience, drawing inspiration from the user-centric design of popular services in the UAE market. In my unique dual role as both the Product Designer and Mobile Developer, I was entrusted with the entire product lifecycle. This meant I was responsible for guiding the project from its initial conceptualization and user flow design all the way to a fully functional, deployed mobile application that served a community of over 150 beta users.

Supporting Media: <https://docs.google.com/document/d/1ZQYzbIM-KAbaULRslgePKtgbJMajCh6I-RoQDcKlfCU/edit?usp=sharing>

My Core Contributions:

- **Product Design (UI/UX):** I designed the entire user journey, starting with user research and creating wireframes, then moving to high-fidelity mockups and interactive prototypes. My focus was on creating a clean, intuitive interface that simplified the car rental process.
- **Mobile Development:** I built the application from the ground up using React Native, ensuring a consistent and responsive experience on both iOS and Android. I implemented the full frontend, connected it to backend APIs, and managed the application's state.
- **User-Centric Optimization:** The application was tested by over 150+ beta users. I was responsible for gathering their feedback to identify pain points and continuously optimize the user experience, leading to a more refined and user-friendly final product.

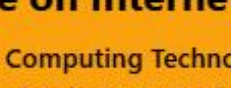
Impact

- Successfully designed and delivered a complete, cross-platform MVP that was validated by over 150 beta users².
- My integrated role as designer and developer streamlined the development process, allowing for rapid iteration and a final product where the technical implementation was perfectly aligned with the design vision.

What I Learned

This project was a deep dive into the critical interplay between product design and mobile development. I learned how to make design decisions that were technically feasible and how to build user interfaces that were both beautiful and performant. Managing the entire process from wireframe to deployed app taught me how to think like a complete product owner, balancing user needs with technical constraints to create a successful product.







5th International Conference on Internet of Things, ICIoT 2025

Department of Computing Technologies
SRM Institute of Science and Technology, Kattankulathur, Chennai, India

APRIL 2nd - 4th, 2025

Paper Submission Deadline: 20th March 2025, Acceptance Notification: 22nd Mar 2025, Final Paper Submission: 25th Mar 2025, Registration Deadline: 26th Mar 2025, Workshop Dates: 3-4 Apr 2025.

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REGISTRATION

Registration Fees

CATEGORY	REGISTRATION AMOUNT
Indian Student/Research Scholar (INR)	₹11000
Academician	₹12000
Industrial Participants	₹13000
Attendee	₹2000
Foreign Research Scholar/(Scholar outside India) (USD)	\$250
Foreign Academician (USD)	\$300

*There will be no refund for registration fees. The publication only includes presented papers.

Register Now

Important Dates

Conference Date: April 2nd to 4th

Paper Submission Deadline:
20th March 2025

Acceptance Notification:
22nd March 2025

Final Paper Submission:
25th March 2025

Registration Deadline:
26th March 2025

Acknowledgment

The Microsoft CMT service was used for managing the peer-reviewing process for this conference. This service was provided for free by Microsoft and they bore all expenses, including costs for Azure cloud services as well as for software development and support.

Registration Details

Authors whose paper has been selected for presentation will be communicated with the details for making online payment. They need to

Project Type: Professional Project (Official platform for an international academic conference)

My Role: Lead Web Developer

Project Description

As the Lead Web Developer, I led a team of 4 to architect and build the official digital platform for the 5th International Conference on Internet of Things (ICIOT 2025). This was a mission-critical project requiring a robust, scalable, and high-performance website to serve as the central hub for a global audience of academics and industry professionals.

My Core Contributions

I built the platform using a modern tech stack (Next.js and a headless CMS) to ensure optimal performance and a seamless content management experience. A key part of my role was successfully integrating the Microsoft Conference Management Toolkit (CMT) to handle the entire academic workflow of paper submissions and peer reviews.

Impact

The platform was a complete success, flawlessly serving over 2,500 global attendees. It provided a reliable and professional hub for registration, submissions, and information dissemination, directly contributing to the smooth operation of a major international academic event hosted by SRMIST

What I Learned

I led a team of 4 talented developers to deliver this enterprise-grade application under strict deadlines for a professional client. This project taught me how to architect for high availability, manage complex third-party integrations, and coordinate cross-functional efforts effectively.

Although the project may appear simple on the surface, its impact and scale make it a significant achievement. Building a mission-critical platform that served over 2,500 global attendees with zero downtime is no small feat—it required meticulous planning, robust engineering, and strong leadership.

This experience reinforced that simplicity in user experience and design often masks deep technical complexity and demands rigorous execution to succeed.

Supporting Media: https://docs.google.com/document/d/1eDiJzDOPPPCif26LYB3bsun44AZfNJCZuVovciCuu_4/edit?usp=sharing